

What a reading proposal should state in order to be assessable: a minimum protocol for undeciphered scripts

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Abstract

Proposals for reading undeciphered scripts appear faster than the field can evaluate them, and the reason is not a shortage of expertise but the absence of a shared minimum of what a proposal must state to be assessable at all. Drawing on the experience of applying a single procedure to five corpora with known or partly known answers (Etruscan, Eteocypriot, Proto-Elamite, proto-cuneiform and Iberian) and to one that has none (Linear A), and on our own withdrawn results, this paper proposes ten requirements: a null model for every distributional claim; a positive control in a corpus with a known answer; a prediction about unseen text; a declaration of how many tests were run, with correction for multiple comparisons; verification of the data in the primary edition; publication of negatives with their figures; an account of the ending of any sign-group identified with a word of another language, or an explicit statement that it is unaccounted for; and, before any absence is interpreted, a check that the instrument could have seen what is said to be missing. None of the ten requires computational methods, and none demands that a proposal be correct: they demand that it be evaluable. For each requirement a case is given in which its absence produced a false positive, and twelve of those cases are the author's own, all withdrawn with their figures. Two further failure modes are described that no null model catches, because they are properties of how a test is built or of how an edition is ordered rather than of the data.

Keywords: undeciphered scripts, method, null models, reproducibility, Linear A, Indus script.

1. The problem

Reading proposals appear faster than the field can evaluate them, and the million-dollar prize for the Indus script (2025) has accelerated the pace. The problem is not a lack of competence: it is that there is no common minimum of what a proposal must declare for another to judge it. Without that minimum, discussion is settled by authority or by exhaustion, and a proposal that is wrong for a demonstrable reason survives beside one that is wrong for reasons nobody can articulate.

This paper proposes no method and evaluates no one else's proposal. It proposes ten requirements, all elementary in other empirical sciences, and for each shows a case in which its absence produced a false positive. Twelve of the cases are the author's own, and they are there deliberately: whoever proposes a protocol should first show where he has failed it.

2. The eight requirements

2.1 A null model for every distributional claim

Every claim of the type “this word occurs with that one”, “this affix is productive”, “these two document classes differ” needs a permutation that preserves what the claim does not purport to explain and destroys only what it does. Any null will not do: the wrong null produces positives.

Own case: we measured the ascendancy-to-capacity ratio of the document-word network at Haghia Triada and at Susa, and both fell within Ulanowicz’s “window of vitality”, an attractive result. With a null preserving row and column sums, the null reproduces the observed value to three decimal places. The index was measuring the marginal distributions. Withdrawn.

Own case 2: we measured the compositionality of the Minoan lexicon (whether long words decompose into recurrent elements) and obtained 0.12 against 0.02 for the shuffled-syllable null, above Greek at equal size. With a null preserving syllable transitions, the observed value falls within the null ($p = 0.60$): it was the phonotactics of the syllabary. Withdrawn.

Rule: the null must preserve everything that is not the hypothesis. A second part must be added to this requirement, which applying the protocol to someone else’s proposal showed to be necessary. Declaring the null is not enough: one must declare **which claim it covers and which it does not**. A null over a set does not cover its members, and this is a common case: in the companion paper on toponyms, fifteen cases jointly exceed the null and none stands on its own, which is stated there expressly. And a null on coincidence, however well built, measures how improbable the observed would be were there no effect; it does not turn a proposal into a certainty, nor does it eliminate apophenia, the bias of seeing structure where there is none, which no calculation internal to the material can rule out. The gap between what a null measures and what is concluded from it is as real a failure as not running one.

2.2 A positive control in a corpus with a known answer

Before applying an instrument to the unknown, one must check that it recovers the known in a comparable corpus. If it does not, the instrument’s silence is not information.

Our use: we calibrated on Etruscan (it recovers the genitive before *clan*, the chronology of syncope and the geography of the sibilants), on Eteocypriot against Cypriot Greek (it separates the two languages by their endings), on Proto-Elamite (it recovers Dahl’s classification of the numerical systems, the institutional heading and the name/commodity partition) and on Iberian (it recovers the formulaic elements).

Counterexample from the field: most reading proposals for Linear A and for the Indus script report no positive control, so it is not known whether their method would see the answer if it were in front of them.

2.3 A prediction about unseen text

A reading that only explains the texts already known cannot be evaluated. The proposal must say what it expects to find in the next inscription and what would refute it.

Our use: the reading of the KU-PA family predicts that its members will go on appearing with the cyperus logogram and with dry commodities, that a new roundel with KA-KU-PA will carry that logogram or a ligature of it, and that the family will not appear as the recipient of liquids. The protocol for what will be done with the Knossos sceptre when it is published is pre-registered before the text is known.

Note: the prediction does not require that the text appear; it requires that it be written down beforehand.

2.4 How many tests were run, with correction for multiple comparisons

A procedure that tests forty hypotheses finds two below 0.05 by chance. The proposal must declare the denominator and correct within each family of tests.

Own case: of eleven suffixes tested in Linear A, three exceed the threshold uncorrected and none survives Bonferroni; we report it as such, and the result we sustain is not “these three suffixes are productive” but “there is a positional signal of morphology that none of the suffixes individually demonstrates”.

Counterexample from the field: screens for lexical coincidence between an unread script and an extensive dictionary, which are in effect thousands of simultaneous tests, almost never declare that number.

2.5 Verify the datum in the primary edition

Derived transliterations, including excellent ones, introduce errors that propagate.

Own case: one of our supports for reading KU-PA as a commodity was a ligature, *304+PA-KU-PA, on a Haghia Triada nodule. The palaeographic database reads three consecutive signs there: the ligature does not exist. It sustained an argument for two days.

Own case 2: the same source transcribes some fraction signs as whole numbers, which made the arithmetic of several tablets fail.

Rule: every datum sustaining a claim is checked in the edition or in the palaeographic database, and the version used is declared.

2.6 Publish negatives with their figures

A negative without a number is not information: “we found no co-exclusion” may mean that there is none or that the corpus is small. With a number, the negative is a measure of power and saves the next investigator the work.

Our use: there is no measurable co-exclusion between Linear A words (zero pairs with expectation above one in 224 documents), and there is in Proto-Elamite with 830 tablets (13 pairs); a network descriptor has no test-retest reliability at these sizes; comparison by sound does not discriminate between candidate languages (all between 1.3 and 1.7 over their own null) either in Linear A or in Linear B, where the same rule produces 383 false positives.

2.7 Account for the ending, or state that it is unaccounted for

When a proposal identifies a sign-group with a known word of another language, the comparison usually rests on the stem and leaves the ending untreated. That is where most false positives enter, because a stem of two or three syllables coincides easily and an unexplained ending may be concealing morphology, a particle, or a misplaced word boundary.

The requirement is cheap: **state what the ending is proposed to be, or state expressly that it is unaccounted for**. It does not demand being right; it demands not being silent.

An own case illustrates it, which is why it is included here. In the companion paper on names formed on plant words we proposed KU-MI-NA-QE as formed on the word for cumin (Mycenaean ku-mi-no, Greek κύμινον, a Semitic loan), on the basis of the shared stem and of its distribution on a sealed roundel, where the sign-group is the party and the logogram the commodity. The vowel -a against Mycenaean -o is no objection, since it follows the correspondence rule measured over 46 pairs (the Minoan final vowel becomes -o in Greek and -a is never preserved). **But the ending -QE was left untreated**, and a consultation with J. M. Jiménez Delgado (University of Seville) identified it as the problem with the case.

Checked in the corpus, the objection holds. There are seven sign-groups ending in -qe among 782 types (0.9%), and only two have their stem attested separately; against the shuffled-syllable null preserving lengths, those two pairs give $p = 0.086$, so -qe does not hold as a productive suffix, unlike -ja, which does with seven pairs. And KA-PA-QE is telling: its stem KA-PA is a Cretan toponym (Uchitel 2002-2003) heading six Haghia Triada tablets, so that KA-PA-QE reads naturally as “and KA-PA”. In Mycenaean, -qe is the copulative enclitic. If Minoan had a comparable enclitic, KU-MI-NA-QE would be the word ku-mi-na with a particle attached rather than a name, and its two contexts admit that reading.

The case is not refuted; it is left **open**: it satisfies two of the three conditions, with the third in dispute. Writing it that way is what this requirement asks, and it is what the original proposal did not do.

2.8 Before interpreting an absence, check that the instrument could have seen it

A zero is the most tempting observation in a small corpus and the easiest to misread. When something does not appear, there are two possible explanations and only one is interesting: that it is absent because the object did not have it, or that it is absent because the instrument could not see it. The requirement consists in ruling out the second before asserting the first.

Two own examples illustrate it, both detected on applying this requirement retroactively.

First. We claimed that no Linear A sign-group ends in -so, and contrasted this with the fact that Cretan toponyms end in -so in Greek (Knossos, Tyliossos, Amnisos). Measuring the full distribution of syllables, it turns out that **the syllable so never occurs in the corpus, in any position**, and with it seven others of the -o column (do, jo, mo, no, qo, wo, zo). The word-final absence was not a phonotactic feature of Minoan: it was a gap in the transliterated syllabary. The claim was withdrawn and replaced by the comparison of each syllable with its own general

frequency, which does measure a positional preference and does not depend on which signs have an assigned value.

Second. Applying by semantic field the functional criterion of a companion paper, names formed on animal words gave zero matches against nine for the plant field. The immediate reading was that Minoan onomastics did not form names on animals, which would be a cultural feature with content. But the search is necessarily made with Mycenaean stems, and in Crete livestock is always written with a logogram and never with syllables: the Minoan words for animals with which to search do not exist. The zero measured the absence of available vocabulary, not the absence of the pattern.

The check is cheap in both cases and consists of the same question: what would have to be in the corpus for the instrument to detect what I say is missing? If that prior condition is not met, the absence is not a datum about the language.

The requirement is therefore stated thus: **every claim of the form “X does not occur” must be accompanied by the check that X could have occurred**, that is, that the sign, word or category was available to the instrument in the material and the transliteration used. Where that check cannot be made, the absence is reported as a property of the available corpus and not of the language.

2.9 When one script is compared with another, the control must be a script of the same kind, and more than one

A rate measured on a script is a property of the script's orthographic conventions as much as of the language behind it. A single comparison witness cannot separate the two.

The case that forced this requirement, obtained and withdrawn within seven hours. Word-final m-syllables occur in 7.1% of Linear A units against 3.1% in Mycenaean Greek, $p = 1.5 \cdot 10^{-7}$. Mimation is a diagnostic feature of Semitic, and this was the only measurement in a long project that favoured a Semitic classification of Minoan. The Cypriot syllabary writing Greek gives **6.2%** over 693 words. Two syllabaries writing languages without mimation give 3.1% and 6.2%, because Linear B usually omits a final consonant while the Cypriot syllabary writes it with a dead vowel. Linear A at 7.1% is indistinguishable from the Cypriot rate. The effect was real; the inference was not.

2.10 Declare the scope of a reading: which archive, which support, which site

A function fixed on one archive is a reading of that archive until it is shown to transfer.

Our case. An inventory of functions was built on Haghia Triada, which is 64% of the Linear A corpus, and read as a description of “the Minoan receipt”. Measured against its own base rates, the second largest series is a different thing: **Khania never sums** (0 of 105 tablets carry a summation term against 41 of 205 at Haghia Triada, $p = 7 \cdot 10^{-11}$), records in fractions, and has no headers or qualifiers of its own. The general claim had to be rewritten as a claim about one archive.

2 bis. Two failure modes that no null model catches

The ten requirements above concern what a proposal must declare. These two concern how a test can be wrong while satisfying all ten, and both were encountered in a single day.

Circularity in the construction of the test. “The unit preceding a logogram never takes a figure” gives 0 of 100 and $p = 3 \cdot 10^{-40}$. It means nothing: if the next token is a logogram then by construction it is not a figure. The null model is correct, the arithmetic is correct, and the result is empty.

Artefacts of the edition’s ordering. A search for sealing batches produced 225 consecutive nodules marked *301, 151 with KU, 149 with KA — apparently the trace of large sealing operations. Of the 854 single-sign nodules of Haghia Triada, **825 consecutive pairs share a sign where 135 would be expected if they were mixed**: the standard edition orders them by their mark. Any analysis of sequence, neighbourhood or batch over that material inherits the ordering, and the edition does not flag it.

Both share a signature worth stating as a rule: **an effect much larger than the phenomenon could plausibly produce is a reason to look for the artefact, not a result**. A p-value of 10^{-40} in a corpus of a few hundred documents should prompt a search for the construction that produced it.

2 ter. What the requirements cost, measured on ourselves

A protocol that nobody applies to themselves is an exhortation. Over one project on Linear A, applying the ten requirements produced **fifteen closed avenues and twelve withdrawn claims**, each filed with its measurement and its reason. Three examples of what the requirements removed:

- A network index placing two Bronze Age archives inside Ulanowicz’s “window of vitality”: the null preserving row and column sums reproduced the observed value to three decimals. It was measuring the marginals.
- A compositionality figure for the Minoan lexicon of 0.12 against 0.02 for shuffled syllables, above Greek at equal size: with a null preserving syllable transitions, the observed value falls inside the null ($p = 0.60$). It was the phonotactics of the syllabary.
- The mimation result of §2.9, which survived seven hours.

None of the twelve was caught by a referee. All were caught by the requirement that the author himself had written down.

3. What the protocol does not require

It does not require computational methods: a proposal made by hand can satisfy all eight requirements, and many computational proposals satisfy none. It does not require giving up intuition, which is the origin of almost every good reading; it requires separating the moment of intuition from the moment of checking. And it does not require that a proposal be correct in order to be publishable: it requires that it be evaluable.

3 bis. The protocol applied to two proposals by others, and calibrated on a language with an anchor

The protocol has been applied to two readings of Linear A that are not ours, without naming their authors, because what matters is the pattern and not the signature.

The first is a decipherment text with seven claims. One partly meets the requirements; two are measurable and fall when measured: the power law offered as proof of “natural language” is obeyed better by the shuffled corpus (R^2 0.95) than by the real one (0.81), and the word breaks presented as successes are right 88.5% of the time against a base rate of 86.5%. The other four state no null and can have none.

The second is a public repository generated with a language model that declares Linear A “solved in five minutes” with 92% confidence. Against the corpus: it gives 1,450 inscriptions with Knossos 450 and Haghia Triada 300 (the corpus has 1,720, Knossos 59, Haghia Triada 1,108); it reads “YA-NE” as wine with confidence 0.90, and the unit does not exist; it proposes “formulas” (oil + 10 + recipient) that occur four times in 89; and it assigns to bare syllables meanings such as “divine invocation”. The only correct items (KU-RO total, PA-I-TO Phaistos, VIN wine) were already published. It is the extreme case of a proposal that states nothing verifiable, and it shows why two of the eight requirements, verifying the datum in the edition and counting the tests, must precede any null: a null on a unit that does not exist is a calculation without an object.

The same procedure, applied to a language that does have an anchor, serves as calibration. Iberian is read and not understood, and has a partial bilingual, the Latinised names of the Turma Salluitana on the Ascoli bronze. Their elements, written in Iberian orthography (which does not distinguish voiced from voiceless stops: Adingibas is atin-kibas), occur in the Hesperia corpus above a null of strings of equal length in 8 of 23 cases (1.2 expected), five after correction; Untermann’s thirty onomastic formants, in 15 of 24 tested. In Latin orthography, without the correspondence, 1 of 16. The same instrument applied to the only near-bilingual source for Minoan, the six Cretan place names of the Egyptian lists, anchors nothing (1 of 9, and that one an artefact of a frequent skeleton). The difference between the two languages is one of data, not of method: two dozen names against six, and the instrument says so with a p in each case. That is what the protocol asks of any proposal: to state how much anchor it has, measured, before saying what it reads.

3 bis bis. The precedent this protocol is meant to avoid

The need for these requirements is not hypothetical. Rao and colleagues (2009) argued in *Science* that the Indus Valley script encodes a natural language, because its conditional entropy resembles that of languages more closely than that of several non-linguistic systems; Lee, Jonathan and Ziman (2010) published an argument of the same kind for the Pictish symbols. Sproat (2010) refuted both with a larger set of non-linguistic and comparative corpora, showing that none of those measures reliably tells writing from non-writing, and proposed a repetition-based measure that classified the same symbols the other way.

What failed there was not the statistics but the control: the comparison was made against the non-linguistic systems the authors chose, and not against the set that could have brought the claim down. That is requirement 2 of this list, and the case shows its cost: a claim published in the most widely read journal in the world, disputed for years, which a better chosen control would have bounded from the start.

4. Why now, and one case in the open

Three reasons. The volume of proposals has grown with the availability of tools and of prizes. The corpora are digitised, so the eight requirements are now cheap: a null is ten lines of code. And the field has produced, in recent years, examples of how it is done well: the testing against nulls of the tripartite division of Cypro-Minoan, the validation of Ithaca with an experiment among historians, and explicit warnings about the discriminating power of coincidences in small corpora. This paper only orders what those works already practise.

In August 2026 a complete Semitic reading of Linear A appeared as a preprint with a DOI, an auditable catalogue of 67 readings, and — unusually, and to its credit — its phonological conversion rules declared explicitly. Declaring the rules is what makes a proposal testable, and the test is instructive for this protocol. Under those rules, 89.1% of Linear A units find a Semitic comparandum, Mycenaean Greek finds one 79.7% of the time, and Packard's fictitious grids (1974) reach 97.1%. Simulated annealing over the free signs finds grids scoring 183 against 75 for the standard grid, changing 37 of 38 signs.

That is requirement 2.2 in action: without a positive control, a criterion that prefers false assignments looks like evidence. The proposal is not refuted by this, and nothing here bears on the comparative Semitic morphology it also rests on. What the measurement fixes is where the weight of the argument must lie, which is the whole purpose of a protocol.

5. Where these requirements come from, and who else has met them

None of the ten is original. Each is standard practice in a field that has had to formalise it, and several have been met in Aegean studies already.

Positive controls and calibration. Corazza, Tamburini, Valério and Ferrara (2022) reclassified the Cypro-Minoan writing system with unsupervised deep learning and validated the method on scripts with known answers before applying it to the unknown one. Assael and colleagues (2022) built their restoration model for Greek epigraphy the same way. Tamburini (2025) surveys the computational approaches and is explicit about which reproduce and which do not.

Verification in the primary edition. Godart and Olivier (1976-1985) is the reference against which every transliteration in this work was checked, and Montecchi (2009) is the study that established how fractions and calculation errors behave in the Linear A documentation. Where the machine-readable transcriptions in circulation disagree with GORILA, this work follows GORILA and says so.

Declaring what a null covers. Duhoux (1978) analysed Linear A linguistically without claiming more than the distribution supported, and his later insistence that resemblance between isolated words proves nothing is the clearest statement in the field of requirement 1.2.8.

Scope and genre. Salgarella and Judson set out the limits of palaeography as a criterion; Davis (2025) surveys what linguistics-based work has and has not established. Steele's work on the Aegean scripts is the standing example of treating writing system and language as separable questions.

And the requirement no one has met. Not one of the decipherment proposals surveyed in the companion paper reports how many tests were run before the one that was published. That is requirement 1.2.4, it is elementary, and its absence is why a century of proposals cannot be compared with one another.

Declaration of assistance

The analysis of the cases and the draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

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